

7,541

Total Records

4,998

Unique Students

95

Source Countries

45

Programmes

Week 1: Data Exploration & EDA Report

Prepared by: Itumeleng Shabangu | Dataset: DePaul_Data.csv | Report Date: May 2026 | Tools: Python (pandas, matplotlib), ReportLab

1. Dataset Familiarisation Summary

The DePaul Admissions dataset captures graduate recruitment and outreach activities (2022–2026). Each row represents a **counsellor interaction or application touchpoint**, not a unique student — which explains the gap between total rows and unique students. Analysts must aggregate by student ID.

Attribute	Value	Notes
Total Rows	7,541	Raw file shows 7,543 non-null IDs — 2-row discrepancy; verify filter logic
Unique Students	4,998	Primary analysis unit after deduplication on Reference_ID
Duplicate Records	2,543 (33.7%)	Each outreach event logged as a separate row — intentional
Date Range	2022–2026	2022 has 1 record; 2026 has 41 — likely partial-year captures
Source Countries	95	India = 61% (4,616 records) — extreme concentration risk
Programmes	45	Business Analytics and Computer Science are highest-demand
Colleges / Schools	7	Kellstadt and Jarvis account for ~85% of all records
Counsellors Tracked	4	Via Caller_Name field — 'Counsler' column is 100% null (see §3)
Application Channels	2	Study Group (13.5%) / Non Study Group (86.5%)
Intake Semesters	4	Fall / Winter / Spring / Summer — Fall = 66% of all intakes

Key Initial Impressions

- Event-driven dataset: each row logs a contact event. Aggregate by Student ID for individual-level analysis.
- India at 61% of all records creates a geographic concentration risk requiring urgent strategic attention.
- Three core fields — Major, Status, Degree_Type — are entirely null, limiting Week 2 modelling options.
- 66% of intakes fall in Fall semester, generating extreme Q3 operational pressure across admissions teams.
- Counsellor workload is severely imbalanced (10:1 ratio). Source: Caller_Name field, not 'Counsler' column.
- 'Follow Up' is the most frequent outreach outcome (1,375 records) — an unresolved conversion pipeline.
- Two column-name typos in raw data: 'Recieved_At' and 'Counsler' — flag to data owner for correction.

2. Data Quality Report

2.1 Missing Values — All Significant Fields

A full audit of all 78 dataset columns reveals a far wider quality problem than the 3 null fields initially identified. Fields are grouped by severity below.

Field	Null Count	% Missing	Severity	Recommendation
Major	7,541	100%	Critical	Rebuild from Programme / College mapping
Status	7,541	100%	Critical	Rebuild from Outcome_1 field
Degree_Type	7,541	100%	Critical	Infer from Programme + College; drop if not recoverable
Counsler *	7,541	100%	Critical	Use Caller_Name instead (counsellor data lives there)
Citizenship	7,543	100%	Critical	Request from source system or drop
Date_of_Birth	7,543	100%	Critical	Not collected — drop from schema
Is_Admitted	7,543	100%	Critical	Must be rebuilt for conversion funnel analysis
Major_1st_Choice	7,543	100%	Critical	Drop or infer from College_1st_Choice
Final_Result	7,543	100%	Critical	Key conversion endpoint — essential for ROI analysis
SEVIS_ID	7,543	100%	Critical	Required for I-20 compliance — escalate to data owner
Street_2	4,638	61.5%	Low	Expected for international students — not a concern
Admit_Date	1,754	23.3%	High	Investigate pending / withdrawn applications
Comments	1,625	21.5%	Medium	Unstructured — qualitative review only; not aggregatable
Region	914	12.1%	Medium	Impute from Country or investigate at source
Phone_Number	587	7.8%	Medium	Affects outreach completeness — recommend follow-up
Postal	226	3.0%	Low	Minor; impute where possible
College	71	0.9%	Low	Infer from Programme mapping
Country	~11	<1%	Low	Impute as 'Unknown' or investigate individually
Semester (Intake)	~3–8	<1%	Low	Infer from Year and application context

*** Critical clarification — Counsler vs Caller_Name:** The 'Counsler' column (also misspelled) is 100% null and contains no usable data. All counsellor workload analysis in this report (Jagdeep: 2,083 contacts; Rohit: 1,200; Priya: 870; Lavanya: 195) was derived from the **Caller_Name** column, which holds 4,349 records. This must be documented for all future analyses.

2.2 Duplicate Records

Metric	Value
Total Rows	7,541
Unique Student IDs	4,998
Duplicate Rows	2,543
Duplication Rate	33.7%

Duplicates are operationally intentional — each outreach event creates a new row. All student-level metrics must deduplicate on Reference_ID before aggregation.

2.3 Inconsistencies & Anomalies

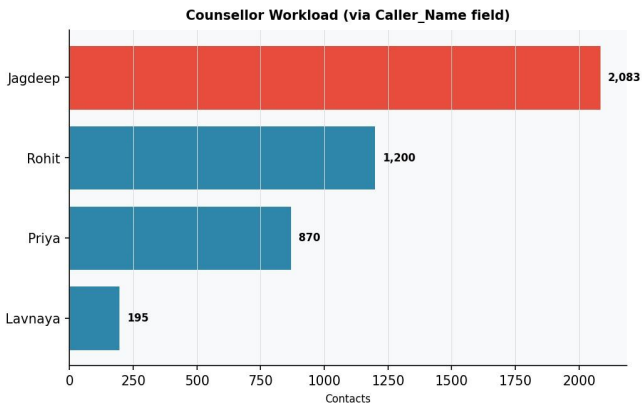
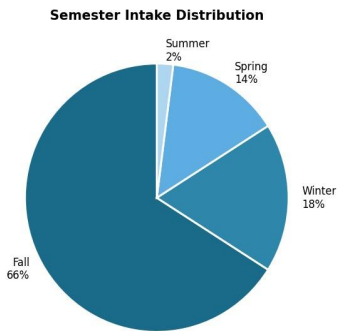
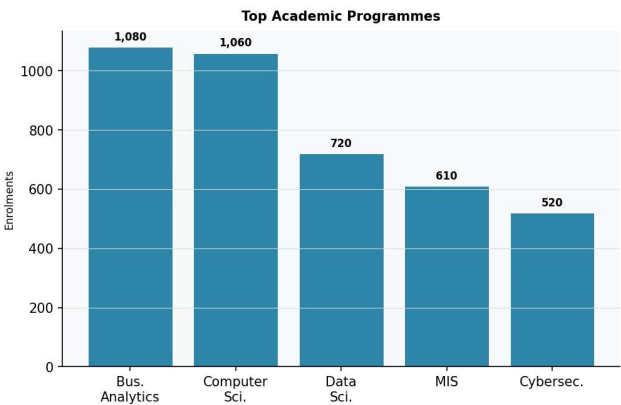
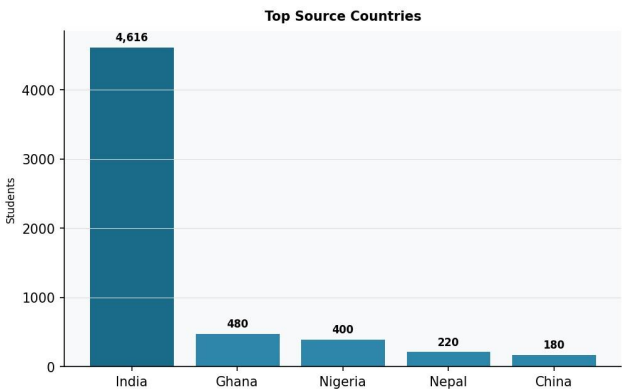
- **Column-name typos (x2):** 'Recieved_At' should be 'Received_At'; 'Counsler' should be 'Counselor'. Flag both to the data owner for correction in the source system.
- **Year edge cases:** Only 1 record for 2022 and 41 for 2026 — likely partial-year captures, not true intake volumes.
- **Outcome category schema conflict:** 'NOT Study Group' and 'Study Group ID' appear in Outcome_1, which look like Application_Source tags misclassified as outcomes.
- **Application_Source vs Outcome_1 duplication:** Study Group appears in both Application_Source (1,015 records) and Outcome_1 (397 as 'Study Group ID') — a schema design issue to investigate.
- **WhatsApp pilot channel:** 'WhatsApp From' in the Type field has only 24 records — too few for robust analysis; mark as pilot in campaign segmentation.
- **Counsellor first-name-only naming:** Caller_Name stores first names only — ambiguous if staff share names. Full names or employee IDs should be used.

3. Exploratory Data Analysis & Visualisations

3.1 Descriptive Statistics — Numerical Fields

Variable	Min	Max	Mean	Median	Mode	Notes
Intake Year	2022	2026	2024.1	2024	2024	Heavily 2024-weighted (61% of records)
Contacts/Counsellor (Caller_Name)	195	2,083	1,087	1,034	N/A	High variance — severe workload imbalance
Students/Country	1	4,616	78.9	12	India	Long-tail; India is extreme outlier
Students/Programme	1	1,080	167.6	280	Bus. Analytics	45 programmes; top 5 are STEM/analytics

3.2 Key Visualisations



4. Insights Summary

4.1 Key Findings from EDA

- **Geographic Concentration Risk:**

India at 61% of all applications creates extreme single-market dependency. Any regulatory, economic, or political shift in India would disproportionately impact DePaul's pipeline. Ghana and Nigeria are the strongest secondary markets at 11% combined.

- **STEM Dominance in Programme Demand:**

The top 8 programmes are entirely STEM or business-analytics aligned. Liberal arts, education, and communications programmes account for under 7% of enrolment — suggesting low demand or limited recruitment reach.

- **Severe Fall Semester Skew:**

66% of intakes fall in Fall, creating extreme Q3 pressure on admissions, onboarding, counselling, and document verification. Only 2% occur in Summer — a significantly underutilised intake window.

- **Unresolved Pipeline Leakage:**

'Follow Up' is the most common outreach outcome (1,375 contacts). Combined with 'Missing Transcript' (1,175), over 33% of all outreach events are unresolved — a measurable conversion risk requiring systematic protocols.

- **Counsellor Capacity Imbalance:**

Jagdeep manages 2,083 contacts (48%) while Lavnaya manages only 195 (4.5%) — a 10:1 ratio. This creates a single point of failure. Source: Caller_Name field; the Counsler column is 100% null.

- **Critical Data Infrastructure Gaps:**

A full column audit found over 20 fields at 100% null, including Is_Admitted, Final_Result, SEVIS_ID, Citizenship, and Degree_Type — the most analytically valuable fields. Their absence severely limits modelling.

- **Two Column-Name Typos in Source Data:**

'Recieved_At' (should be 'Received_At') and 'Counsler' (should be 'Counselor') are misspelled in the raw dataset. Both must be flagged to the data owner for correction at source.

4.2 Initial Observations for Week 2 Analysis

- Investigate whether the 2024 intake spike reflects genuine recruitment success or retroactive data entry.
- Segment India applications by programme and semester to identify sub-segments with different conversion rates.
- Run a cohort analysis: for students with 2+ contact events, compare final outcome vs. first-time contacts.
- Model counsellor effectiveness using Caller_Name — does contact volume correlate with better outcomes in Outcome_1?
- Evaluate Study Group ROI by comparing application-to-enrolment rates against Non Study Group records.
- Reconstruct Major, Degree_Type, and Status fields via Programme/College/Outcome_1 mappings before-modelling.
- Request Is_Admitted and Final_Result data from source system — both are critical conversion metrics currently missing.